ColRadPy

Theory Manual

Collisional radiative kinetics and spectral emission

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This manual develops the rate equations, the elimination of excited states, generalized collisional radiative coefficients, line emission, time-dependent and charge-state solutions, and the energy-balance extension. Equations are mapped to the routines that implement them. Blue notes explain implementation choices; orange notes identify current limitations or qualifications. The worked checks execute the development working tree; file hashes and the recorded HEAD are given in `docs/source/theory/figures/provenance.json`.

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1 Purpose and model scope

ColRadPy converts level-resolved atomic data into populations, photon emission coefficients, and effective rates for charge-state evolution. The basic input is an ADAS ADF04 description of one ion: level energies, radiative probabilities, electron-impact collision strengths, and any supplied parent-resolved ionization, recombination, or charge-exchange data. ColRadPy implements the generalized collisional radiative (GCR) approach [1, 9]. Statements about the implementation refer to the source used to generate the figures and checks of this manual [12].

The plasma parameters are prescribed. Electrons are Maxwellian at a specified temperature, atomic processes are represented by rate coefficients, and emitted photons escape. Ground and metastable populations can be retained as independent drivers. The principal calculation does not solve radiative transfer, a spatial transport equation, or the electron distribution function. Its rate coefficients can instead be passed to a transport or plasma model.

The collisional radiative regime lies between two useful limits. At sufficiently low density, excitation is balanced mainly by radiative decay, giving the coronal limit. At high density, reversible collisions tend toward a Boltzmann distribution within a charge stage; ionization and three-body recombination also tend toward Saha balance when their inverse processes and statistical weights are consistent. These are limits to test, not assumptions that every fusion or astrophysical plasma automatically satisfies.

1.1 Two calculations with different meanings

The quasistatic calculation sets derivatives of selected excited populations to zero while retaining the ground and metastables as drivers. It produces linear response coefficients. The fully time-dependent calculation propagates the entire supplied rate matrix from an initial vector. A separate ionization-balance calculation joins effective coefficients from adjacent charge stages. Keeping these distinctions prevents an excited-state response per parent from being mistaken for a normalized ion fraction.

1.2 Calculation flow

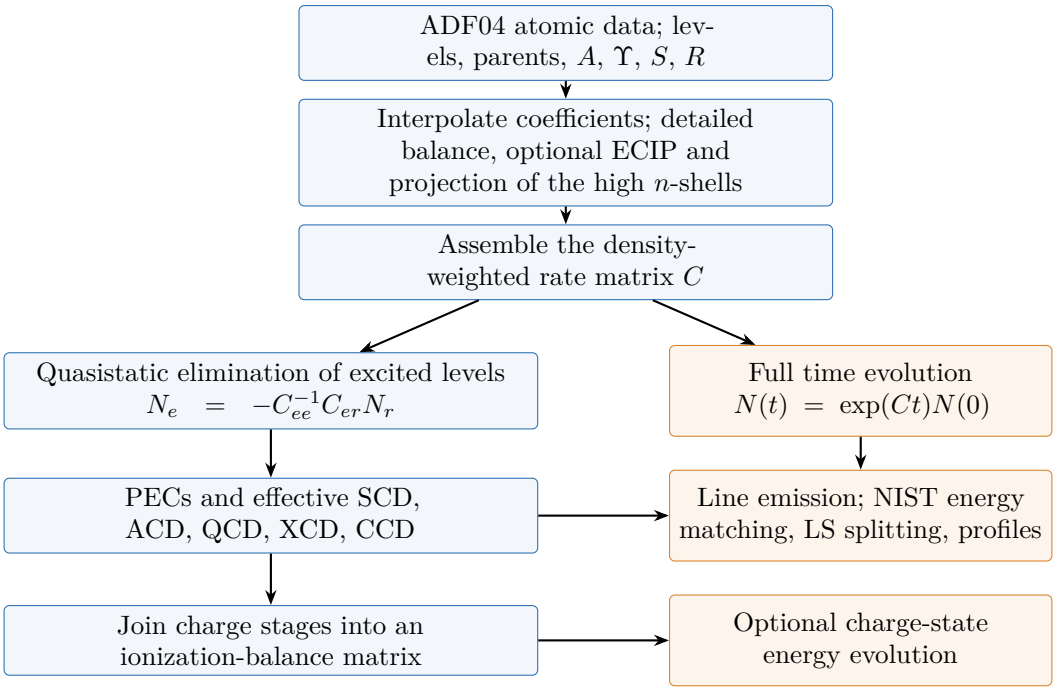


Figure 1: The calculation paths are related but do not share the same population normalization. The full matrix can also include parent or reservoir states; their interpretation must be specified.

2 Notation, indexing and units

Let $N_i^{(z)}$ denote the number density of level i in charge state z. Indices i, j generally label bound levels; ρ, σ label retained ground or metastable levels; p, q label parents in charge state $z+1$. The subscripts e and r on matrix blocks denote eliminated excited and retained states, respectively. They do not denote electron density.

The defining matrix convention is

$$\frac{\mathrm{d}N_i}{\mathrm{d}t} = \sum_j C_{ij} N_j, \qquad C_{ij} : \; j \longrightarrow i. \tag{2.1}$$

Thus a column is a source state and a row is a destination. Off-diagonal rates are nonnegative for a physical model. This convention differs from some stored coefficient arrays, as listed in Section 15.1.

The conversions needed repeatedly are

$$\Delta E_{\mathrm{eV}} = \frac{\Delta E_{\mathrm{cm}^{-1}}}{8065.54}, \qquad T_{\mathrm{K}} \simeq 11604.5\, T_{\mathrm{eV}}, \tag{2.2}$$

$$\lambda_{\mathrm{vac,nm}} = \frac{10^7}{|E_u - E_\ell|_{\mathrm{cm}^{-1}}}, \qquad n_{\mathrm{m}^{-3}} = 10^6 n_{\mathrm{cm}^{-3}}. \tag{2.3}$$

The code uses several slightly different rounded energy-conversion constants. For reproducibility, the exact expressions in a routine take precedence over a claim that all those constants are identical.

ADF04 level and parent identifiers are one-based. Python level indices are zero-based; `metas=[0,2]` retains the first and third bound levels. The complement is then $[1, 3, 4, \ldots]$, not the contiguous interval beginning at the number of metastables. A parent identifier is also not interchangeable with the position at which that identifier was discovered in a file.

Quantity	Meaning	Units
T_e, T_i	Temperature represented as an energy	eV
$E_i,\ I_p$	Tabulated level energies and parent thresholds	cm^{-1}
n_e, n_h, N_i	Electron, neutral-donor and level densities	cm^{-3}
$A_{u\ell}$	Spontaneous emission probability	s^{-1}
q, S, α^{RR+DR}, K	Two-body coefficients	$cm^3\,s^{-1}$
$\alpha^{(3)}$	Three-body recombination coefficient	$cm^6\,s^{-1}$
C	Density-weighted rate matrix	s^{-1}
Υ	Effective collision strength	dimensionless
λ	Internal wavelength	nm
PEC	Photon emission coefficient	$cm^3\,s^{-1}$
SCD, ACD, QCD, XCD, CCD	Effective two-body-form coefficients	$cm^3\,s^{-1}$

Table 1: Core conventions. Effective coefficients may themselves depend on density. The energy-transfer helper functions use SI densities and masses.

3 Atomic data and rate coefficients

3.1 Level structure and statistical weights

The ADF04 reader stores configurations, multiplicities, orbital labels, energies, and a weight-related field `atomic['w']`. The matrix builder uses $g_i = 2w_i + 1$. In an LS-resolved file the level weight is $(2S+1)(2L+1)$; in a resolved-J description it is $2J+1$. The parent term must be interpreted consistently with that resolution. The header provides parent ionization thresholds, while level annotations describe parentage channels. They need not enumerate identical sets of observed pathways [3].

Parent weights. parent_statistical_weight parses LS and resolved-J parent terms, including fractional J, and uses $g_p = (2S+1)(2L+1)$ or $g_p = 2J+1$ as appropriate. Header parent counts and discovered pathway IDs still require care: an entry can have explicit ionization data even when its parentage annotations are incomplete.

3.2 Maxwellian electron-impact excitation

For an electron-energy distribution $f_E(E)$ normalized to unity,

$$q_{\ell u} = \int_{\Delta E}^{\infty} \sigma_{\ell u}(E) \sqrt{\frac{2E}{m_e}}\, f_E(E)\, \mathrm{d}E. \tag{3.1}$$

The tabulated effective collision strength $\Upsilon_{\ell u}(T)$ packages the Maxwellian average in a convenient dimensionless form. The implemented rate conversion for $u > \ell$ is

$$q_{u\ell}(T_e) = \frac{2.1716 \times 10^{-8}}{g_u} \sqrt{\frac{13.6058}{T_e}}\, \Upsilon_{\ell u}(T_e), \tag{3.2}$$

$$q_{\ell u}(T_e) = \frac{g_u}{g_\ell} \exp\left(-\frac{\Delta E}{T_e}\right) q_{u\ell}(T_e). \tag{3.3}$$

Here ΔE and T_e have the same energy units. Both coefficients are in $cm^3\,s^{-1}$. Their ratio enforces collisional detailed balance: $N_u/N_\ell = (g_u/g_\ell)\exp(-\Delta E/T_e)$ when collisions dominate and no other mechanism fixes the ratio.

Implementation. make_electron_excitation_rates interpolates Υ onto the user grid. populate_cr_matrix constructs the upward and downward rates and multiplies them by electron density. The excitation, ionization and three-body Boltzmann factors convert cm^{-1} to eV with hc/e (1/8065.544), so excitation alone drives the populations to the Boltzmann distribution at high density; the ECIP exponent keeps the constants of r8ecip.

3.3 Spontaneous decay, ionization and recombination

For a radiative transition $u \to \ell$, the population-transfer rate is $A_{u\ell}N_u$, with no electron-density multiplier. The total radiative loss is $\Gamma_u = \sum_{\ell<u} A_{u\ell}$.

Let S_{ip} denote ionization from bound level i to parent p. The reduced ionization quantity in the supported ADF04 convention is converted as

$$S_{ip}(T_\mathrm{e}) = \widetilde{S}_{ip}(T_\mathrm{e}) \exp[-\chi_{ip}/T_\mathrm{e}], \qquad \chi_{ip} = I_p - E_i. \tag{3.4}$$

An interpolated reduced coefficient is not itself the final ionization rate. The input parentage factor can be applied through `scale_file_ioniz`, which multiplies the interpolated reduced coefficients whatever the interpolation; the ionization balance passes the option to each stage. Some files hold placeholders instead of rates at their lowest temperatures, where the rate underflowed when the file was made (reduced ionization rates of 10^{-2} to $10^{40}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$ in OpenADAS neon files). Reduced ionization rates above $10^{-6}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$ and recombination rates above $10^{-7}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$, far above physical values, are left out with a warning, and those temperatures take the rates from the other points of the transition. Used literally, they reach three-body recombination through detailed balance. Smaller artifacts stay below these bounds: values at the lowest temperatures that fall towards the next ones faster than any rate of the kind, such as a recombination coefficient of 2.6×10^{-8} followed by $10^{-30}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$ in OpenADAS `cop98#10_ls#ne1`. Recombination falls at most as $T^{-3/2}$ (radiative recombination into levels bound by less than kT, dielectronic recombination above its resonances), and reduced ionization rises as $T^{1/2}$ near threshold and falls as about $T^{-1/2}$ far above it. Points at the lowest temperatures that fall faster than T^{-2} (recombination) or T^{-1} (reduced ionization) towards a later point are left out with a warning, from the lowest up to the first that does not.

Radiative and dielectronic recombination supplied by a file are collected in $\alpha_{ip}^{\mathrm{RR+DR}}$. The effective recombination coefficient used at a particular density is

$$R_{ip}(T_\mathrm{e}, n_\mathrm{e}) = \alpha_{ip}^{\mathrm{RR+DR}}(T_\mathrm{e}) + n_\mathrm{e}\alpha_{ip}^{(3)}(T_\mathrm{e}). \tag{3.5}$$

Its contribution to a matrix element is $n_\mathrm{e}R_{ip}$, so the three-body term enters with two factors of electron density. Zero or missing file data must be distinguished from a reliable physical assertion that a process is absent.

ECIP is an optional approximate completion of missing ionization channels. ecip_rates prepares the allowed parent channels, the levels marked $\{p\}$ in the level list that lie below the ionization potential of parent p (also above the first ionization potential), and passes Kelvin temperatures to r8ecip. suppliment_with_ecip chooses channels using parent annotations and zero entries in the assembled ionization array. This is an approximation to missing atomic data, not a substitute for a convergence study or a higher-quality calculated rate.

3.4 Three-body recombination and Saha detailed balance

Equilibrium of $i + e \rightleftharpoons p + e + e$ requires

$$N_i^* n_\mathrm{e} S_{ip} = N_p^* n_\mathrm{e}^2 \alpha_{ip}^{(3)}. \tag{3.6}$$

Using the electron translational partition factor gives

$$\frac{\alpha_{ip}^{(3)}}{S_{ip}} = \frac{g_i}{g_p}\frac{h^3}{2(2\pi m_e k_B T_\mathrm{K})^{3/2}}\exp\left(\frac{\chi_{ip}}{T_\mathrm{e}}\right) \simeq \frac{g_i}{g_p}\frac{1.6564\times10^{-22}}{T_\mathrm{e}^{3/2}}\exp\left(\frac{\chi_{ip}}{T_\mathrm{e}}\right)\ \mathrm{cm}^3. \tag{3.7}$$

The last numerical form takes T_e in eV and includes the free-electron spin factor of two. The implementation uses the close approximation 1.656742×10^{-22}.

Stable low-temperature evaluation. Three-body recombination uses the reduced ionization coefficient directly, avoiding a product of an underflowed ionization rate and an overflowing inverse exponential. This gives a finite positive result at a fixed positive temperature when the reduced coefficient is finite and positive; it does not establish a universal $T \to 0$ limit.

Standalone LTE helpers. boltzmann_population returns ground-relative level populations or total-normalized fractions. saha_population includes both stage partition functions, ionization energy, electron mass, and electron density; it returns N_{z+1}/N_z or the two-stage ionized fraction. Energies are in cm^{-1}, temperature in eV, and electron density in cm^{-3}.

3.5 Thermal charge exchange

A donor species at temperature T_h and density n_h can capture an electron into bound level i from parent p. Writing its coefficient $K_{ip}(T_h)$, the gain is $n_\mathrm{h}K_{ip}N_p^\mathrm{CX}$. The density multiplier is the donor density, not automatically the electron density. The routine make_cx_rates_from_file interpolates the donor-temperature grid. The user-supplied donor and electron grids must be compatible with the selected grid or paired layout.

4 Interpolation and asymptotic continuation

The interpolation option parser separates a leading `log` or `loglog` flag from the SciPy interpolation kind. The `log` flag selects *logarithmic temperature coordinates*; it does not take the logarithm of the rate coefficient. For example,

$$\widehat{r}(T_\mathrm{e}) = \mathcal{I}[\ln T_k, r_k]\,(\ln T_\mathrm{e}) \tag{4.1}$$

still interpolates the ordinates r_k directly. Log-temperature interpolation alone therefore does not guarantee positive extrapolated rates: the extrapolation is clipped at zero, which the Be I recombination and ionization coefficients reached a few times above the file temperatures.

By default (`loglog_slinear`), recombination and reduced ionization coefficients are interpolated as

$$\widehat{r}(T_\mathrm{e}) = \exp\mathcal{I}[\ln T_k, \ln r_k]\,(\ln T_\mathrm{e}), \tag{4.2}$$

over the positive values of each transition. Between two temperatures this gives the geometric mean of their values where linear interpolation of r_k gives the arithmetic mean, which is never smaller and overestimates coefficients that change by orders of magnitude between neighbouring temperatures (by up to 45 times for OpenADAS Ne^{7+} and 236 times for Be I recombination). Outside the file temperatures the coefficients follow a power law through the two nearest values, $\widehat{r} \propto T^{\,s}$, with the exponent limited to physical values: $-3/2 \le s \le 0$ for recombination, so it neither grows nor falls faster than radiative and dielectronic recombination; for reduced ionization

$-1 \le s \le 1/2$ above the file temperatures and $s \ge 0$ below them, where a falling coefficient is held constant. The options without `loglog` keep the interpolation of the coefficients themselves; excitation and charge exchange use them, and excitation is continued outside the file temperatures as described below.

The Burgess–Tully helpers compress an unbounded temperature coordinate into $0 \le x < 1$. With $\theta = k_B T/\Delta E$ and a positive scaling parameter c (default 1.5), the four implemented transforms are An available infinite-energy point constrains the transformed high-temperature

Type	$x(\theta)$	Transformed ordinate y
1	$1 - \ln c/\ln(\theta + c)$	$\Upsilon/\ln(\theta + \mathrm{e})$
2	$\theta/(\theta + c)$	Υ
3	$\theta/(\theta + c)$	$(1 + \theta)\Upsilon$
4	$1 - \ln c/\ln(\theta + c)$	$\Upsilon/\ln(\theta + c)$

Table 2: Coordinate and ordinate definitions in `burgess_tully_rates.py`. The original routines use Kelvin and wavenumbers to form θ.

continuation. Otherwise the code fits the final calculated points in transformed space. The inverse transform is necessary to recover Υ. The end of a calculation is a limit of the dataset, not of ColRadPy: the Mo I collision data end near 17 eV and are continued above it with the Burgess–Tully transforms [1]. The infinite-energy information constrains a continuation but does not validate an extrapolation far from the calculated range.

Continuation. The caller applies all four high-temperature categories and uses one aligned fallback slot per category. Below the file grid it applies a nonnegative transformed continuation or retains the first collision strength for transitions whose data start far above threshold. This numerical policy prevents negative rates but does not establish atomic-data accuracy outside the calculated range.

5 Atomic data from the Flexible Atomic Code

The Flexible Atomic Code (FAC) [13] computes level energies, radiative rates and cross-sections of electron collisions in a jj-coupled basis. `read_fac` reads the ASCII tables that its `PrintTable` writes with energies in eV into the ADF04 dictionary of Section 3, so the matrix of Section 6 is built as for an ADF04 file and the solver is unchanged. The levels of the ion are sorted by energy; those of the next ion are the parents, with I_p measured from the ion's ground. FAC levels have an exact J but no LS term: their L and S are -1, and the parent weights of Eq. (3.7) are $2J_p + 1$.

5.1 Maxwellian averages from each threshold

A process with threshold E_0 has the rate coefficient

$$q(T) = \int_{E_0}^{\infty} \sigma(E)\, v(E)\, f(E,T)\, \mathrm{d}E = e^{-E_0/T}\, T \int_0^{\infty} \sigma(E)\, v(E)\, \tilde{f}(E,T)\, e^{-u}\, \mathrm{d}u, \qquad E = E_0 + uT, \tag{5.1}$$

with $\tilde{f} = f\, e^{E/T}$. The distribution is Maxwell–Jüttner,

$$f(E,T) = \frac{\gamma^2 \beta\, e^{-\gamma/\theta}}{\theta K_2(1/\theta)\, mc^2}, \qquad \gamma = 1 + \frac{E}{mc^2}, \quad \theta = \frac{T}{mc^2}, \quad v = c\beta, \tag{5.2}$$

which is $2\sqrt{E/\pi}\,T^{-3/2}e^{-E/T}$ for $T \ll mc^2$. The last integral of Eq. (5.1) is taken with 64 Gauss–Laguerre nodes, so the nodes start at the threshold and follow the Boltzmann factor whatever E_0/T, and $\tilde{f}$ uses the exponentially scaled Bessel function, so nothing underflows. The reduced coefficient $q\,e^{E_0/T}$ gives the effective collision strength $\Upsilon = q\,e^{\Delta E/T} g_\ell \sqrt{T/13.6058}/2.1716 \times 10^{-8}$, with the constants of Eq. (3.2) so that the matrix holds the integrated rate, and the reduced ionization rate of Eq. (3.4). The cross-sections follow from FAC's tables as FAC defines them, with $k^2 = 2E(1+\alpha^2 E/2)$ and E in hartree: $\sigma = \pi a_0^2 \Omega/(g_\ell k^2)$ for excitation, $a_0^2\Omega/(gk^2)$ for ionization, and, from the bound-free oscillator strength density S, $\sigma_{\mathrm{RR}} = 2\pi\alpha S a_0^2 (\alpha E_\gamma)^2/[2\varepsilon(1+\alpha^2\varepsilon/2) g_p]$ for radiative recombination of a free electron of energy ε with a photon E_γ.

Between the tabulated energies Ω and S are interpolated log-log. Below the first one Ω of excitation and S are held, as they are finite at threshold, and Ω of ionization follows FAC's fit. Above the last, excitation joins FAC's Bethe–Born limit $\Omega = b\ln(E/E_0) + c$ at the energy FAC computed it for (FAC's manual advises against its direct values at hundreds of thresholds), and a transition without a Born limit (exchange) follows the power law of its last two values, at least as steep as E^{-1}, as FAC does; ionization follows FAC's fit $\Omega = p_0 \ln x + p_1 y^2 + p_2 y/x + p_3 y/x^2$ ($x = E/I$, $y = 1 - 1/x$), and recombination FAC's fit of S, valid at high energies, with the angular momentum of the captured electron, both scaled to the last tabulated value.

5.2 Autoionization and dielectronic recombination

Dielectronic capture into an autoionizing level d from parent p has the strength S_{DC} ($\mathrm{cm^2\,eV}$) at the resonance energy $E_r = E_d - I_p$, so $\alpha_{\mathrm{DC}} = S_{\mathrm{DC}}\, v(E_r)\, f(E_r, T)$, which for a Maxwellian is the detailed balance of the autoionization rate A^a_{dp}, $\alpha_{\mathrm{DC}} = \frac{g_d}{2g_p}(h^2/2\pi mT)^{3/2} A^a_{dp} e^{-E_r/T}$. ADF04 data have no autoionization channel, so `_fold_autoionization` folds it into the other rates. With $R_{d\ell}$ the radiative rates of d, $R_d = \sum_\ell R_{d\ell}$ and $A^a_d = \sum_p A^a_{dp}$, from the lowest autoionizing level up,

$$b_d = \frac{\sum_\ell R_{d\ell}\, b_\ell}{R_d + A^a_d}, \qquad P_{dp} = \frac{A^a_{dp} + \sum_\ell R_{d\ell} P_{\ell p}}{R_d + A^a_d}, \tag{5.3}$$

with $b = 1$ and $P = 0$ for levels that do not autoionize: b_d is the fraction of the population of d that ends in stable levels and P_{dp} the fraction that reaches parent p. Every flow into d (capture, excitation and cascades, $R_{d'd} \to b_d R_{d'd}$) is multiplied by b_d, and the excitation of a stable level ℓ into d that autoionizes becomes its ionization (excitation-autoionization),

$$\widetilde{S}_{\ell p} \mathrel{+}= \tilde{q}_{\ell d}\, P_{dp}\, e^{-(E_d - I_p)/T}, \tag{5.4}$$

with $\tilde{q}_{\ell d}$ the reduced excitation rate. The population of d is then the inflow over $R_d + A^a_d$, as with the autoionization in the matrix, while collisions from d are slow beside its decays (below about $10^{20}\,\mathrm{cm^{-3}}$ for highly charged ions), and so are its lines, the dielectronic recombination and the ionization. Left out are the capture that autoionizes to another parent, a process of the next ion; the emission of lines between two autoionizing levels beyond b of the lower; and autoionizing levels without radiative decays, which never stabilize. The two-photon decays of H-like $2s$ and He-like $1s2s\,{}^1S_0$ [15] add to their A-values.

Checks. The tests write FAC tables for a toy ion and check: a constant Ω gives $\Upsilon = \Omega$; $\Omega = b\ln x + c$ gives $c + b\,e^y E_1(y)$ with $y = E_0/T$; a constant S gives the closed-form radiative recombination; capture follows the detailed balance above; and the folded rates give the populations, SCD and ACD of the model with the autoionization in its matrix to 10^{-14}. On FAC's own reference tables the cross-sections made from Ω and S are those FAC prints to 8×10^{-5}. The average of Mewe's Gaunt-factor cross-section [14] for $\mathrm{Ar^{17+}}$ $1s$–$2p_{1/2}$ matches its closed form to 5×10^{-5} from 10 eV to 85 keV, and the relativistic average is lower by 2% at 10 keV and 17% at 85 keV.

The averages assume a thermal distribution: ColRadPy derives de-excitation and three-body recombination by detailed balance, which holds for no other. Magnetic-sublevel tables are not read, and FAC parents have no LS term to link the stages of an ionization balance.

6 The collisional radiative matrix

For bound levels $i \neq j$, define $q_{j\to i}$ as the corresponding electron-impact coefficient. Then

$$C_{ij} = A_{j\to i} + n_\mathrm{e} q_{j\to i}, \tag{6.1}$$

$$C_{ii} = -\sum_{j\neq i}(A_{i\to j} + n_\mathrm{e} q_{i\to j}) - n_\mathrm{e}\sum_p S_{ip}. \tag{6.2}$$

Here forbidden or upward spontaneous transitions have $A = 0$. For adjacent-charge parents,

$$C_{pi} = n_\mathrm{e} S_{ip}, \qquad C_{ip} = n_\mathrm{e} R_{ip}. \tag{6.3}$$

The corresponding parent diagonal loses recombination into the lower stage. A closed collection of states satisfies

$$\sum_i C_{ij} = 0 \quad \text{for every source column } j, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\sum_i N_i = 0. \tag{6.4}$$

Column sums need not vanish in a deliberately open subsystem; the missing population must then be accounted for as an external source or sink.

Implementation. populate_cr_matrix stores the main generator in `cr_matrix['cr']`; `A_ji` is stored with upper-level index first, so radiative entries are transposed when inserted into the generator. `cr_red` later contains only the selected excited-state block. The separate `cr_loss` array holds only the level-loss diagonal of that generator.

The CX driving columns used by the quasistatic solution have no matching losses: they attribute emission to CX from a given parent population. solve_time_dependent instead moves the CX gains to the recombining-parent columns and subtracts them from the parent diagonal, so CX depletes the evolving parent and the transient generator satisfies Eq. (6.4). The CX columns are then empty; their initial populations and sources must be zero.

The off-diagonal sign, column sums, and inverse-channel density powers are useful checks before any matrix is solved. A linear algebra routine can solve a matrix with negative or incorrectly scaled atomic rates without identifying the underlying physical error.

7 Projection of the high n-shells

The levels of an ADF04 file usually cover the lowest principal shells, $n \leq n_1$. The levels of higher shells, up to the series limits of the parents, are missing from the matrix of Section 6, yet most recombination passes through them: radiative capture into high n, dielectronic capture into Rydberg states, and the cascades that follow. At high density they also open stepwise ionization. For neon, the effective recombination into Ne^{3+}, Ne^{4+} and Ne^{5+} from the ADAS `cop98#10` ADF04 files alone is 0.17–0.69 of the ADAS coefficients made from the same files with the high shells included (5–100 eV), and the SCD of Ne^{2+} at 1 eV and $10^{14}\,\mathrm{cm}^{-3}$ is 0.007 of the ADAS one.

7.1 Condensed bundle-nS couplings

A bundle-nS calculation (ADAS204) groups the levels built on a parent ρ of the next ion with spin system S into n-shell populations x_n, from the lowest shell to the series limit, and eliminates all but its lowest shells. The condensed couplings are given for each parent and spin system on a temperature–density grid (ADF17 files), in the convention of losses positive:

$$\sum_m P_{nm}\, x_m = n_\mathrm{e} N_\rho\, r_n, \tag{7.1}$$

where P_{nm} (s^{-1}) is the coupling from shell m to shell n (P_{nn} the total loss of shell n), r_n ($\mathrm{cm}^3\,\mathrm{s}^{-1}$) the recombination into shell n by all routes, $I_{n\rho'}$ (s^{-1}) the ionization of shell n to parent ρ', and $Q_{\rho'}$ ($\mathrm{cm}^3\,\mathrm{s}^{-1}$) the recombination from ρ that ionizes to another parent ρ' without reaching the lowest shells. The same files give the direct parts: the shell couplings D_{nm}, direct ionization $D^\mathrm{I}_{n\rho'}$ and direct radiative, dielectronic, three-body and charge-exchange recombination d_n.

Shells u above the highest shell of the ADF04 levels are eliminated from Eq. (7.1) onto the kept shells l (a Schur complement):

$$P' = P_{ll} - P_{lu}P_{uu}^{-1}P_{ul}, \qquad r' = r_l - P_{lu}P_{uu}^{-1}r_u, \tag{7.2}$$

$$I' = I_l - \left(P_{uu}^{-1}P_{ul}\right)^\mathsf{T} I_u, \qquad Q' = Q + I_u^\mathsf{T} P_{uu}^{-1} r_u, \tag{7.3}$$

and the direct transfers from kept to eliminated shells become indirect. The ADF04 file already has the direct couplings between its levels, so they are subtracted. The indirect couplings through the high shells are then

$$K = P' - D' - \mathrm{diag}\Big(\sum_{\rho'} D^\mathrm{I}\Big), \quad I^\mathrm{ind} = I' - D^\mathrm{I}, \quad r^\mathrm{ind} = r' - d, \quad Q^\mathrm{ind} = Q'. \tag{7.4}$$

Negative values up to 10^{-3} of the projected and direct values they are the difference of are round-off (the ADF17 files have 11 digits) and are set to zero. Larger ones show condensed data that are not reliable, as at the lowest temperatures of most ADF17 files; those temperatures are left out of that coupling of the parent and spin system.

7.2 Expansion over the levels

An ADF18 expansion file (`a17_p208` kind) assigns each ADF04 level i to a parent, spin system and shell $n(i)$, with a weight w_i normalised over the levels of that shell (statistical weights). Level populations add up to shell populations, so a rate per unit shell population applies to each of its levels, while a flow into a shell is shared by its levels with their weights:

$$R^\mathrm{ind}_{i\rho} = w_i\, r^\mathrm{ind}_{n(i)}, \qquad S^\mathrm{ind}_{i\rho} = I^\mathrm{ind}_{n(i)\rho}/n_\mathrm{e}, \tag{7.5}$$

$$G_{ij} = -w_i\, K_{n(i)n(j)} \quad (n(i) \neq n(j)), \qquad X^\mathrm{ind}_{\rho'\rho} = \sum_S Q^\mathrm{ind}_{\rho'}. \tag{7.6}$$

Transfers within a shell are not described by shell populations and are omitted. These terms are added to the matrix of Section 6: G_{ij} to C_{ij}, $n_\mathrm{e}S^\mathrm{ind}_{i\rho}$ to $C_{\rho i}$ and $n_\mathrm{e}R^\mathrm{ind}_{i\rho}$ to $C_{i\rho}$, each with its loss on the diagonal of the source, so Eq. (6.4) still holds. The indirect parent coupling X^ind is added to XCD (Section 9.4). Populations, PECs, SCD, ACD, QCD and XCD then include the high shells.

Below the first temperature T_0 of a coupling c on the ADF17 grid (without the temperatures left out), it continues with the Boltzmann factor of its threshold ΔE:

$$c(T) = c(T_0)\Big(\frac{T}{T_0}\Big)^a \exp\Big[-\Delta E\Big(\frac{1}{kT} - \frac{1}{kT_0}\Big)\Big], \tag{7.7}$$

with a the slope of $\ln(c\,e^{\Delta E/kT})$ against $\ln T$ between its two lowest temperatures, limited as for the ADF04 rates (Section 4): between -1.5 and 0 for recombination and the parent coupling, at least 0 for the reduced ionization and transfers. Every indirect path out of a level starts with its excitation into the high shells, so the threshold of $S^{\mathrm{ind}}_{i\rho}$ is the ionization energy χ_i of level i; that of G_{ij} is $\chi_j - \bar{\chi}_{n_1} n_1^2/(n_1+1)^2$, the energy to reach the shell above the highest kept one, n_1, whose levels have the mean ionization energy $\bar{\chi}_{n_1}$; that of $X^{\mathrm{ind}}_{\rho'\rho}$ is the energy between the parents when ρ' lies above ρ; and recombination has none. Above the grid the values at its edge are kept.

Choices of the implementation. Ionization through the shells of one parent to another parent is the autoionization of core-excited Rydberg states, part of excitation-autoionization, which the ionization rates of the ADF04 files (ADF07 data) already include. It is left out. ADAS208 also credits only the group's own parent but keeps that loss on the diagonal; ColRadPy omits both, so the matrix conserves particles. The diagonal follows from the expanded gains and ionization rather than from K_{nn}; the two agree when the condensed data conserve particles. The condensed data are interpolated linearly in $\ln T_\mathrm{e}$, $\ln n_\mathrm{e}$ and the logarithm of each coupling, without the temperatures where it is not reliable, and continued below the ADF17 grid by Eq. (7.7), whose threshold for the transfers scales the binding energy of the highest kept shell hydrogenically; both give a warning when the user temperatures need them. The fractional-parentage corrections of the lowest shell (switches `IEFPRE`, `IEFPRS`, set only by the superseded `cbnm93` files) are not implemented, and indirect charge-exchange recombination is not represented.

Routines and data. read_adf17 reads the condensed matrices and read_adf18 the expansion files; projection.project forms and expands the indirect couplings. `colradpy(..., projection=(adf17, adf18))` stores them in `rates['projection']` (make_projection, called by solve_cr for each solve on its temperatures and densities, with the files read once) and populate_cr_matrix adds them to the generator. In an ionization balance, `projection` takes one pair, or `None`, per ADF04 file. The ADF18 file names the ADF04 and ADF17 files it maps; its level numbers follow the order of the ADF04 file, before any sorting by energy. Levels it maps to spin systems or shells that the ADF17 file does not have get no projection, with a warning. The recombination rates of the ADF04 file are zero-density data; the density dependence of dielectronic recombination through the high shells (the collisional redistribution of doubly excited states, Eq. 45 of Summers et al. [9]) reaches ColRadPy only through the ADF17 data.

8 Quasistatic elimination of excited levels

Partition the population vector into retained states N_r and excited states N_e. The full system is

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} N_r \\ N_e \end{pmatrix} = \begin{pmatrix} C_{rr} & C_{re} \\ C_{er} & C_{ee} \end{pmatrix}\begin{pmatrix} N_r \\ N_e \end{pmatrix}. \tag{8.1}$$

When excited-state relaxation is fast compared with evolution of the drivers,

$$0 = C_{ee}N_e + C_{er}N_r, \qquad N_e = -C_{ee}^{-1}C_{er}N_r. \tag{8.2}$$

This requires an invertible, relaxing excited block. Including a long-lived state in the eliminated set can invalidate the approximation even if the numerical inverse exists. Time-scale separation should be assessed from the excited-block decay rates and the time scale of the retained dynamics.

Substitution into the retained equations gives the Schur complement

$$C_{\mathrm{eff}} = C_{rr} - C_{re}C_{ee}^{-1}C_{er}. \tag{8.3}$$

This formula is the common origin of the effective coefficients below. For a closed model, correctly eliminating excited states preserves column-sum conservation in the retained equations.

8.1 Choosing retained levels and testing time-scale separation

Summers et al. [9] order the intrinsic lifetimes of an ion of charge z as $\tau_a \ll \tau_o \ll \tau_m$: auto-ionizing levels ($\tau_a \sim 10^{-13}\,\mathrm{s}$), ordinary excited levels ($\tau_o \sim 10^{-8}/z^4\,\mathrm{s}$) and metastables ($\tau_m \sim 10/z^8\,\mathrm{s}$, radiative decay). For most fusion and astrophysical plasmas $\tau_{\mathrm{plasma}} \sim \tau_g \sim \tau_m \gg \tau_o \gg \tau_{e\text{-}e}$, with τ_g the relaxation time of the ground populations by ionization and recombination and $\tau_{e\text{-}e}$ the thermalization time of the free electrons. This ordering underlies the quasistatic picture: the ground and metastable populations evolve with the plasma and are retained, while the excited populations and the free electrons (a Maxwellian distribution) relax to them. Redistribution of the excited populations by collisions, on a time τ_{red}, defines critical densities, $\tau_{\mathrm{red}} \sim \tau_o$ and $\tau_{\mathrm{red}} \sim \tau_a$, which separate low levels, high singly excited levels and doubly excited levels; the projection of Section 7 treats the high levels.

A spectroscopically metastable level and a level that must be retained dynamically are not the same thing [1]. A weak spontaneous decay can produce a large population, but collisions and ionization can shorten its actual relaxation time. Conversely, a collection of ordinary excited levels can participate in a slow, strongly coupled mode. The retained set is therefore a modeling choice for the density, temperature, initial condition and diagnostic being considered.

For constant matrix blocks, write the instantaneous quasistatic response as $N_e^{\mathrm{qs}}(t) = -C_{ee}^{-1} C_{er} N_r(t)$ and let $\delta = N_e - N_e^{\mathrm{qs}}$. The exact lag equation is

$$\dot{\delta} = C_{ee}\delta - \dot{N}_e^{\mathrm{qs}}, \tag{8.4}$$

$$\delta(t) = \mathrm{e}^{C_{ee}t}\delta(0) - \int_0^t \mathrm{e}^{C_{ee}(t-u)} \dot{N}_e^{\mathrm{qs}}(u)\,\mathrm{d}u. \tag{8.5}$$

There are two distinct errors: an initial excited-state relaxation transient and the lag behind a changing driver. After the first decays, a small ratio of excited relaxation time to driver-evolution time supports the approximation. The decaying eigenvalues and eigenvectors of C_{ee} help assess this, although a nonnormal matrix can amplify transients and a gap in eigenvalues alone is not a complete error bound. Changing plasma conditions also changes the response matrix itself and contributes to $\dot{N}_e^{\mathrm{qs}}$.

A useful practical test is to start full and reduced models from consistent retained populations, vary the retained set, and compare the predicted line emission over the required observation times. A ground-only quasistatic model eliminates every other bound state, including levels that might need their own evolving populations. A large excited population identifies a candidate for examination, but a small population can still be an important feeder of a diagnostic line. Eigenmodes are combinations of levels; each eigenvalue should not be labeled as the lifetime of one level without examining its eigenvector.

Comparing the full time-dependent model, which evolves every level, with reductions that retain selected ground and metastable populations and eliminate the other levels quasistatically shows that the adequacy of a retained set depends on the species and conditions (Table 3) [1]:

Published case	Plasma conditions	Reported observation
Be I	$T_{\mathrm{e}} = 20$ eV, $n_{\mathrm{e}} = 10^{13}$ cm^{-3}	Retaining the important metastable substantially improves the time-dependent line-emission prediction over a ground-only reduction.
Mo I	$T_{\mathrm{e}} = 8$ eV, $n_{\mathrm{e}} = 5 \times 10^{11}$ cm^{-3}	Even a ten-metastable reduction does not reproduce the full time-dependent line history.

Table 3: Retained sets compared with the full time-dependent model [1], not rerun for this manual. They do not set a universal number of retained levels for either species.

The Be example exhibits a separation between slowly relaxing modes and the faster excited manifold. The Mo example has a more gradual spectrum of relaxation rates, making a small retained set less successful, which is what the lag and emission-convergence tests above detect. A ground-only reduction also loses the ability to represent an independently evolving metastable fraction.

8.2 Separate excitation, recombination and CX drivers

The excited-state forcing can be separated as

$$0 = C_{ee}N_e + C_{em}N_m + n_{\mathrm{e}}R_{ep}N_p + n_{\mathrm{h}}K_{ep}N_p^{\mathrm{CX}}. \tag{8.6}$$

Define the dimensionless response matrices

$$F = -C_{ee}^{-1}C_{em}, \qquad G = -C_{ee}^{-1}n_{\mathrm{e}}R_{ep}, \qquad H = -C_{ee}^{-1}n_{\mathrm{h}}K_{ep}, \tag{8.7}$$
$$N_e = FN_m + GN_p + HN_p^{\mathrm{CX}}.$$

Each column states a response to one independently prescribed driving population. These are not probabilities whose sums must all be one.

Implementation. solve_quasi_static constructs `beta`$= -C_{er}$ and computes `pops`$= C_{ee}^{-1}$`beta`. The excitation columns come first, followed by recombination and then CX columns when enabled. `processed['F']` denotes an excitation response without the additional $1/n_{\mathrm{e}}$ used in some ADAS definitions. Matching symbols without matching their density convention produces unit errors.

CX-only construction. When CX is enabled but both radiative and three-body recombination are disabled, the CX columns still follow the parent columns, which the generator keeps whether or not recombination is used.

8.3 Normalization is a choice of driving population

For a single ground driver, an excitation response $F_i = N_i/N_0$ can be converted to a fraction relative to the ion total:

$$\frac{N_0}{N_{\mathrm{ion}}} = \frac{1}{1+\sum_i F_i}, \qquad \frac{N_i}{N_{\mathrm{ion}}} = \frac{F_i}{1+\sum_j F_j}. \tag{8.8}$$

This transformation changes the denominator of an excitation coefficient. It is not applicable without modification to $G_i = N_i/N_p$, whose parent belongs to another charge state. The independent drivers are explicitly distinguished in GCR emissivity definitions [2].

Default normalization. With one retained metastable, excitation columns are normalized by default; recombination and CX columns remain relative to their independent parents. With more than one retained metastable, excitation normalization defaults off. The `default_pop_norm` option controls this choice. For example, an excited level with $A = 1\,\mathrm{s}^{-1}$ and parent-driven injection $n_{\mathrm{e}}\alpha = 2\,\mathrm{s}^{-1}$ retains the required response $N_e/N_p = 2$.

8.4 Unsummed contributions and dominant feeding mechanisms

The individual terms of the excited-state solution identify the important populating channels [1]. Let a label an excitation, recombination or CX driver, and write $\beta_{ja} = -C_{ja}$ for the negative of

its driving-block entry into eliminated level j. Define

$$U_{ija} = (C_{ee}^{-1})_{ij}\beta_{ja} \quad \text{(no sum over } j\text{)}, \qquad N_i = \sum_a \sum_j U_{ija} N_a. \tag{8.9}$$

The sum over j gives the response of level i to driver a; leaving it unsummed assigns that response to the level where the driver injects population into the eliminated manifold. For a fixed driver with a nonzero total response, the fractions

$$f_{j|i,a} = \frac{U_{ija}}{\sum_k U_{ika}} \tag{8.10}$$

identify the injection channels carrying the largest contribution. This interpretation requires physical nonnegative rates and a relaxing excited block. A common normalization applied to one driver's response cancels in these fractions, but different drivers must be weighted by their actual populations before they are combined.

The stored contribution tensor. The implementation stores U in `processed['pop_lvl']` with axes *destination excited level, injection excited level, driver, temperature, density.* Both excited-level indices refer to the reduced level map, not to physical bound-level numbers. Summing the second axis reconstructs `processed['pops']` under the same normalization. Paired mode replaces the last two axes by one pair index.

An entry with $j = i$ represents direct injection into the observed upper level, followed by its full response through C_{ee}^{-1}. An entry with $j \neq i$ represents injection elsewhere followed by redistribution or a cascade. These are algebraic injection-channel contributions, not exclusive microscopic paths or counts of the last collision: the inverse already contains repeated transfers. For the Mo I lines near 379.8 and 550.6 nm, these terms, with population-weighted metastable contributions, separate strong direct ground injection from feeding through other populated levels [1]. An unweighted PEC column alone gives neither line's contribution to the measured intensity, which needs the weighting of Eq. (10.1).

The same accounting can resolve effective rates. For example, level i contributes $S_{ip}F_{i\rho}$ to indirect ionization, and contributes $C_{\rho i}G_{ip}/n_e$ to capture feeding retained level ρ. Keeping these terms unsummed can guide a reduced atomic model, provided its emission and effective-rate errors are checked afterward. Bundling levels is another way to reduce the dynamical size of heavy-species models [1]; choosing a short list of retained metastables is not by itself a population-preserving bundling, and ColRadPy provides no automatic bundled model.

9 Generalized collisional radiative coefficients

Effective coefficients combine direct processes with paths through eliminated excited states. Their definition is a reduction of the generator, not simply a sum of raw atomic rates. All formulas in this section assume the unnormalized responses of Eq. (8.7) and explicitly prescribed retained populations. The stored arrays and their index order are listed in Section 15.1.

9.1 SCD: effective ionization

For a retained bound level ρ and destination parent p,

$$\mathcal{S}_{\rho p} = S_{\rho p} + \sum_i S_{ip}F_{i\rho}. \tag{9.1}$$

The second term describes excitation or redistribution followed by ionization. Both terms have coefficient units. Its matrix contribution from ρ to p is $n_e\mathcal{S}_{\rho p}$.

9.2 ACD and CCD: effective capture

The capture coefficient from parent p into retained level ρ is

$$\mathcal{A}_{\rho p} = R_{\rho p} - \sum_{ij} C_{\rho i}(C_{ee}^{-1})_{ij} R_{jp}. \tag{9.2}$$

The second term incorporates capture into excited levels followed by their collisional redistribution and radiative cascades. Replacing R by K defines $\mathcal{C}_{\rho p}$, the CCD coefficient. The rate multipliers are n_e and n_h, respectively. In an ionization matrix expressed per electron density, the CCD entry carries $n_\mathrm{h}/n_\mathrm{e}$.

9.3 QCD: redistribution within a retained bound manifold

For two different retained bound levels,

$$\mathcal{Q}_{\sigma\to\rho} = \frac{C_{\rho\sigma} + \sum_i C_{\rho i} F_{i\sigma}}{n_\mathrm{e}}, \qquad \rho \neq \sigma. \tag{9.3}$$

This includes direct mixing and indirect paths through the excited manifold. The diagonal is kept zero in the stored QCD array; its population loss is inserted separately when the charge-state matrix is assembled.

9.4 XCD: redistribution between parent metastables

A parent p can recombine into an excited level of the lower stage and then ionize into a different parent q. The coefficient for this path is

$$\mathcal{X}_{p\to q} = -\sum_{ij} C_{qi}(C_{ee}^{-1})_{ij} R_{jp} = -n_\mathrm{e} \sum_{ij} S_{iq}(C_{ee}^{-1})_{ij} R_{jp}, \qquad p \neq q. \tag{9.4}$$

The factor n_e in the second form is essential. Dimensional analysis gives

$$[n_\mathrm{e} S][C_{ee}^{-1}][R] = (\mathrm{s}^{-1})(\mathrm{s})(\mathrm{cm}^3\,\mathrm{s}^{-1}) = \mathrm{cm}^3\,\mathrm{s}^{-1}. \tag{9.5}$$

This is equivalent to dividing the parent-to-parent off-diagonal block of the Schur complement by electron density. The recombination rows must use the same excited-level ordering as C_{ee}. The same factor is in the definition of XCD of Summers et al. (their Eq. 17) [9].

XCD contraction. The contraction uses $n_\mathrm{e}S$ and selects recombination rows by the actual eliminated-level map, including noncontiguous metastable selections such as $[0, 2]$. The resulting coefficient has units cm^3/s and agrees with the parent block of the direct Schur complement divided by electron density.

10 Derived emission and power coefficients

Spectrum-line and radiated-power coefficients follow from the quasistatic responses of Eq. (8.7), as in Section 1.2.2 of Summers et al. [9]. Their responses $\mathcal{F}_{u\rho}^{(\mathrm{exc})} = F_{u\rho}/n_\mathrm{e}$ and $\mathcal{F}_{up}^{(\mathrm{rec})} = G_{up}/n_\mathrm{e}$ include the electron density that the F and G of ColRadPy leave out (Section 8.2).

10.1 Photon emissivity coefficients

For an optically thin line $u \to \ell$, the angle-integrated photon emissivity divides by driving population, a metastable ρ of the ion or a metastable p of the next ion (the parent):

$$\epsilon_{u\ell} = A_{u\ell} N_u = n_\mathrm{e} \sum_\rho N_\rho \mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho} + n_\mathrm{e} \sum_p N_p \mathrm{PEC}^{\mathrm{rec}}_{u\ell,p} + n_\mathrm{h} \sum_p N_p^{\mathrm{CX}} \mathrm{PEC}^{\mathrm{CX}}_{u\ell,p} \quad [\mathrm{photons\,cm^{-3}\,s^{-1}}]. \tag{10.1}$$

This defines the excitation, recombination and charge-exchange photon emissivity coefficients ($\mathrm{cm^3\,s^{-1}}$),

$$\mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho} = A_{u\ell} F_{u\rho} / n_\mathrm{e}, \tag{10.2}$$

$$\mathrm{PEC}^{\mathrm{rec}}_{u\ell,p} = A_{u\ell} G_{up} / n_\mathrm{e}, \tag{10.3}$$

$$\mathrm{PEC}^{\mathrm{CX}}_{u\ell,p} = A_{u\ell} H_{up} / n_\mathrm{h}. \tag{10.4}$$

The upper level can also be a retained metastable σ. Its population relative to the driver ρ is $\delta_{\sigma\rho}$, so the lines it emits to the levels below it, the forbidden transitions between metastables of Summers et al. (their Eq. 26), have

$$\mathrm{PEC}^{\mathrm{exc}}_{\sigma\ell,\rho} = \delta_{\sigma\rho} A_{\sigma\ell} / n_\mathrm{e}, \qquad \mathrm{PEC}^{\mathrm{rec}}_{\sigma\ell,p} = \mathrm{PEC}^{\mathrm{CX}}_{\sigma\ell,p} = 0. \tag{10.5}$$

With the populations normalized to the whole ion (Section 8.3) an excitation PEC is multiplied by N_ρ / N_ion. This is the coefficient for a stage-to-stage ionization balance, where a metastable treated as an ordinary level can have a quasistatic population comparable to that of the ground [9]. For an isotropic emitter, a line-of-sight photon radiance is $\int \epsilon_{u\ell}\, \mathrm{d}s/(4\pi)$. The 4π factor is an observation convention and is not already part of $A_{u\ell} N_u$. The ionizations per emitted photon, S/XB, are defined from the excitation PECs in Section 10.3.

CX PEC convention. The implementation divides *all* computed driving columns by n_e when forming `processed['pecs']`. Its stored CX columns therefore contain $AH/n_\mathrm{e} = (n_\mathrm{h}/n_\mathrm{e})\mathrm{PEC}^{\mathrm{CX}}$ as defined above. Use electron density with that stored response, or explicitly convert to a donor-normalized coefficient before using Eq. (10.1).

10.2 Radiated power coefficients

Each PEC carries the line power $\Delta E_{u\ell}\,\mathrm{PEC}$, in $\mathrm{W\,cm^3}$ with the photon energy $\Delta E_{u\ell} = hc/\lambda_\mathrm{vac}$ in J ($\times 10^{-6}$ for $\mathrm{W\,m^3}$). Summed over the lines they give, for each driver, the low-level line power coefficient and the recombination power coefficient (Summers et al., their Eqs. 24–25):

$$\mathrm{PLT}_\rho = \sum_{u\ell} \Delta E_{u\ell}\, \mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho} = \sum_{u\in e} \sum_{\ell<u} \frac{\Delta E_{u\ell} A_{u\ell} F_{u\rho}}{n_\mathrm{e}} + \sum_{\ell<\rho} \frac{\Delta E_{\rho\ell} A_{\rho\ell}}{n_\mathrm{e}}, \tag{10.6}$$

$$\mathrm{PRB}_p = \sum_{u\ell} \Delta E_{u\ell}\, \mathrm{PEC}^{\mathrm{rec}}_{u\ell,p} = \sum_{u\in e} \sum_{\ell<u} \frac{\Delta E_{u\ell} A_{u\ell} G_{up}}{n_\mathrm{e}}, \tag{10.7}$$

with $u \in e$ the eliminated levels, and the charge-exchange power PRC_p in the same way from $\mathrm{PEC}^{\mathrm{CX}}$. The last term of Eq. (10.6) is the radiation of the metastable ρ itself, Eq. (10.5). Summers et al. write it as a separate term, ADAS includes it in PLT, and it dominates for a metastable with a fast decay: for the $\mathrm{Ne^0}$ $2\mathrm{p^5 3s\,{}^3P}$ term ($A = 1.6 \times 10^7\,\mathrm{s^{-1}}$) ColRadPy without it gave 0.02 of the ADAS PLT (`plt96r`) at $10^{12}\,\mathrm{cm^{-3}}$, and with it 1.00. Summers et al. include bremsstrahlung in PRB, and the ADAS PRB adds the recombination continuum and the radiation among the high n-shells (Section 15.3); the `prb` of ColRadPy is Eq. (10.7) only, the line power of the levels of the ADF04 file after capture.

Power outputs. `processed['pls']` holds $\Delta E_{u\ell}$ PEC for every line and driver, and `plt`, `prb` and `prc` its sums over the excitation, recombination and CX columns, in $\mathrm{W\,cm^3}$, with the photon energy from the level-energy difference. Every pair of levels is a line, by lower and then upper level as in `pec_levels`, the lines of the retained metastables included. The time-dependent `['td']['pecs']`, `pls` and `plt` use the same lines with the time-dependent populations of all levels, metastables included.

10.3 S/XB: ionizations per emitted photon

The diagnostic ratio S/XB relates ionization to excitation-driven line emission [1]. For a single retained driver ρ, define the total effective ionization coefficient $\mathcal{S}_\rho = \sum_p \mathcal{S}_{\rho p}$. For line $u \to \ell$,

$$(S/XB)_{u\ell,\rho} = \frac{\mathcal{S}_\rho}{\mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho}}. \tag{10.8}$$

This is Eq. 23 of Summers et al. [9], the SCD of the driver summed over the parents over its excitation PEC. Both coefficients use the same electron-density and driving-population convention, so the ratio is dimensionless: ionizations per excitation photon. The numerator can instead be restricted to a specified destination parent, provided that restricted meaning is stated.

For multiple retained populations, form the rate ratio before interpreting it as a diagnostic:

$$(S/XB)_{u\ell}[N_m] = \frac{\sum_\rho N_\rho \mathcal{S}_\rho}{\sum_\rho N_\rho \mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho}}. \tag{10.9}$$

This is not an unweighted average of the individual metastable ratios. Different metastable fractions can produce different ionization-to-photon ratios at the same temperature and density.

If excitation dominates the observed line and S/XB is approximately constant through the emitting column, its photon radiance $I_{u\ell}$ gives

$$\int \dot{n}_{\mathrm{ion}}\,\mathrm{d}s = (S/XB)_{u\ell}\,4\pi I_{u\ell}. \tag{10.10}$$

If the ratio varies spatially, the local ionization rate must instead be integrated with the corresponding emissivity. Interpreting that column ionization rate as a surface erosion or influx rate additionally requires a geometry and transport model: escape without ionization, repeated recombination and ionization, and recombination/CX photons can invalidate a simple one-ionization-per-entering-particle interpretation. A zero excitation PEC does not define a finite ratio.

S/XB output. S/XB is a ratio of stored results and is not stored itself: `get_sxb` returns Eq. (10.8) for every line and retained driver, from the parent sum of `scd` and the matching excitation column of `pecs`, with the same normalization, and NaN where a line has no excitation emission. `write_sxb_adf13` writes it to an ADF13 file [10], as ADAS208 does, an undefined ratio as the format's floor 10^{-74} with a warning; the file needs a temperature–density grid. Both need ionization in the calculation. A per-driver photon rate in $\mathrm{s^{-1}}$ must not be mixed with an ionization coefficient in $\mathrm{cm^3/s}$.

11 Wavelengths, fine structure and line profiles

11.1 Wavelengths and NIST energy matching

Vacuum wavelength follows Eq. (2.2). For a chosen air convention, $\lambda_{\mathrm{air}} = \lambda_{\mathrm{vac}}/n(\lambda_{\mathrm{vac}})$. The air-wavelength helper uses a pressure- and temperature-adjusted dispersion expression:

$$D(\xi) = 8342.54 + \frac{2406147}{130 - \xi^2} + \frac{15998}{38.9 - \xi^2}, \quad \xi = 1000/\lambda_{\mathrm{vac,nm}}, \tag{11.1}$$

$$n = 1 + 10^{-8}\left[D(\xi)\frac{P}{96095.43}\frac{1 + 10^{-8}(0.601 - 0.00972T_C)P}{1 + 0.0036610T_C} - P_w(0.037345 - 0.000401\xi^2)\right]. \tag{11.2}$$

Here P and water-vapor pressure P_w are in Pa and T_C is in Celsius; the defaults are 101325 Pa, 0 Pa and 15 °C. This empirical expression is not a wavelength-independent physical law. It has mathematical poles near 87.706 and 160.334 nm. ColRadPy keeps vacuum wavelengths below 200 nm and applies an air conversion only above that threshold. An optical air convention is thus separate from the vacuum convention used for UV emission.

NIST matching changes line positions by replacing matched level energies. It does not recompute collision strengths or establish that LS splitting reproduces a resolved collisional radiative calculation. Configurations, term labels, J values, and ambiguous repeated labels must be matched consistently. The split structure is constructed when needed before the levels are shifted.

11.2 Statistical splitting of LS multiplets

For a spin-allowed electric-dipole LS transition, relative component strengths can be expressed through an angular factor

$$W_{J_u J_\ell} \propto (2J_u + 1)(2J_\ell + 1)\begin{Bmatrix} L_u & J_u & S \\ J_\ell & L_\ell & 1 \end{Bmatrix}^2. \tag{11.3}$$

Normalize the allowed factors over the multiplet and distribute the term PEC by $\mathrm{PEC}_{J_u J_\ell} = \mathrm{PEC}_{\mathrm{LS}} W / \sum W$. This describes relative LS angular strengths with the associated statistical population assumption. It does not add resolved-J collisional dynamics or configuration mixing. split_pec_multiplet evaluates equivalent Condon–Shortley factors and renormalizes them for spin-conserving dipole multiplets ($|\Delta L| \leq 1$). Intercombination lines and lines with $|\Delta L| > 1$, which these factors do not describe, are kept unsplit at the wavelength between the statistically weighted term centres.

11.3 Doppler and natural profiles

For an emitter of mass m at temperature T_i,

$$\Delta\nu_D = \frac{\nu_0}{c}\sqrt{\frac{2eT_i}{m}}, \qquad \phi_D(\nu) = \frac{\exp[-(\nu - \nu_0)^2/\Delta\nu_D^2]}{\sqrt{\pi}\Delta\nu_D}. \tag{11.4}$$

This is a Gaussian with standard deviation $\Delta\nu_D/\sqrt{2}$ and FWHM $2\sqrt{\ln 2}\,\Delta\nu_D$. The broadening API takes T_i in eV and the emitter mass in kg.

For lifetime broadening of an upper and lower state,

$$\Delta\nu_L = \frac{\Gamma_u + \Gamma_\ell}{2\pi}, \qquad \gamma = \frac{\Delta\nu_L}{2}, \qquad \phi_L(\nu) = \frac{\gamma/\pi}{(\nu - \nu_0)^2 + \gamma^2}. \tag{11.5}$$

The Lorentzian FWHM is $\Delta\nu_L$. Density-dependent pressure broadening requires additional physics; it is not supplied merely by this lifetime width.

For a profile tabulated per wavelength rather than per frequency, preserve area with $\phi_\lambda = \phi_\nu|\mathrm{d}\nu/\mathrm{d}\lambda|$ and consistent length units. The code evaluates a frequency-shaped kernel on a wavelength grid and renormalizes its finite-grid area numerically. That construction is a useful narrow-line profile but should not be assumed identical to an exact Jacobian-transformed wide profile. Grid width and resolution control omitted wings and numerical normalization.

Profile grid resolution. Natural profiles are evaluated on offsets $\Delta\lambda$ from the line centre, stored in `dwave_arr`, with the frequency offsets $\Delta\nu = -c\,\Delta\lambda/[\lambda_0(\lambda_0 + \Delta\lambda)]$. A natural width can be far below the float64 resolution of the wavelength itself (5×10^{-15} nm for the Be I line at 454.8 nm, where float64 resolves 6×10^{-14} nm), and frequencies near 10^{15} Hz are resolved only to 0.1 Hz, so neither absolute grid can carry such profiles. The absolute wavelengths `wave_arr` of these lines coincide.

12 Time-dependent evolution and matrix exponentials

For fixed plasma conditions and an optional constant source s,

$$\dot{N} = CN + s, \qquad N(t) = \mathrm{e}^{Ct}N(0) + \int_0^t \mathrm{e}^{C(t-u)}s\,\mathrm{d}u. \tag{12.1}$$

If $C = V\Lambda V^{-1}$ has a complete eigenbasis, the source-free result is $V\mathrm{e}^{\Lambda t}V^{-1}N(0)$. Each source-driven mode obeys

$$v_k(t) = v_k(0)\mathrm{e}^{\lambda_k t} + c_k\phi(\lambda_k, t), \quad c = V^{-1}s, \quad \phi(\lambda, t) = \begin{cases} (\mathrm{e}^{\lambda t} - 1)/\lambda, & \lambda \neq 0, \\ t, & \lambda = 0. \end{cases} \tag{12.2}$$

For small $|\lambda t|$, evaluating the numerator with `expm1` avoids subtracting nearly equal floating-point numbers. Small $|\lambda|$ by itself does not make a mode negligible: its decay is governed by $|\lambda t|$. An alternative formulation is the augmented exponential

$$\begin{pmatrix} N(t) \\ 1 \end{pmatrix} = \exp\left[t\begin{pmatrix} C & s \\ 0 & 0 \end{pmatrix}\right]\begin{pmatrix} N(0) \\ 1 \end{pmatrix}. \tag{12.3}$$

For constant C, these formulas propagate the initial state at $t = 0$; the first requested output time is not a new integration origin. The initial vector is a physical assumption about the species: atoms leaving a surface start in the ground state, and their emission evolves as metastable populations build up [1]. The rate equations accept any initial vector, and the time to form metastables corresponds to a distance from the surface only with a model of the particle motion and geometry.

12.1 Complex modes and real populations

A real, nonsymmetric rate matrix can have complex conjugate eigenvalues. For example, a conservative three-state cycle has

$$C = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \lambda = 0,\ -\tfrac{3}{2} \pm \tfrac{\sqrt{3}}{2}\,\mathrm{i}. \tag{12.4}$$

The complex modes are legitimate and must be retained in the reconstruction. For real initial data and a real source, their imaginary contributions cancel in the exact populations. Taking

the real part of eigenvectors or eigenvalues changes the dynamics. Taking the real part of a reconstructed population is justified only after a numerical assessment of its imaginary residual. Real-arithmetic matrix-exponential methods avoid those complex intermediates.

12.2 Defective and ill-conditioned eigenbases

Not every valid generator is diagonalizable. An equal-rate chain is a useful counterexample:

$$C = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad N(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad N(t) = \begin{pmatrix} \mathrm{e}^{-t} \\ t\mathrm{e}^{-t} \\ 1 - (1+t)\mathrm{e}^{-t} \end{pmatrix}. \tag{12.5}$$

The $t\mathrm{e}^{-t}$ term comes from a Jordan block. A nominal inverse of numerical eigenvectors can return plausible values that omit this term. Near a defective matrix, eigenvector conditioning matters more than the visual separation of eigenvalues alone. A robust algorithm should assess that conditioning and use a matrix-exponential or equivalent stable method when needed.

Transient algorithms. The transient solvers check eigenvector conditioning and compare selected populations against a matrix exponential. Untrusted points are recomputed with scipy.linalg.expm; a constant source uses an augmented matrix. Real populations are returned, and slow physical modes are retained according to their evolution over the requested times rather than an absolute rate cutoff. The cached evaluator needs the original rate matrix to perform the same exponential fallback.

12.3 Stationary solutions

A normalized stationary population of a closed model satisfies

$$C\pi = 0, \qquad \mathbf{1}^{\mathsf{T}}\pi = 1, \qquad \pi_i \geq 0. \tag{12.6}$$

For a unique equilibrium, one redundant balance equation can be replaced by the normalization equation. Rescaling balance equations helps avoid loss of weak-rate constraints. If multiple disconnected closed classes exist, their relative weights depend on the initial condition; normalization alone does not determine a unique distribution.

A rate matrix that conserves particles (non-negative off-diagonal rates and columns that sum to zero) is solved by Grassmann–Taksar–Heyman elimination, which uses only the off-diagonal rates and has no subtractions, so every population keeps the relative accuracy of the rates however small it is. Solutions through the eigenvectors or a linear solve are accurate only relative to the largest population: for a resolved carbon balance at 10 eV they gave C^{5+} 2×10^{-5} where the exact value is 3×10^{-16}. For other matrices the stationary solver tries the smallest-magnitude eigenmode and checks its residual and normalization; an unreliable projection is replaced by a normalized stationary solve. The ionization-balance wrapper calls this routine directly.

13 Ionization balance across charge stages

Collect all retained metastables across the modeled stages into a vector x. At fixed plasma parameters,

$$\dot{x} = Mx + s, \qquad M = n_{\mathrm{e}}B. \tag{13.1}$$

Here B is a coefficient matrix with units cm^3/s. Its adjacent-stage blocks contain SCD upward and ACD downward. QCD connects retained levels within the lower stage, and XCD contributes

connections within the parent stage through the lower stage. A given stage can therefore receive both its own QCD and XCD contributions computed from the preceding charge state.

For each off-diagonal transition, the same coefficient is subtracted from the source diagonal. Consequently $\mathbf{1}^{\mathsf{T}} B = 0$ in a closed calculation. The CX downward blocks use $(n_{\mathrm{h}}/n_{\mathrm{e}})\mathcal{C}$ inside B, so the final multiplication by electron density recovers the physical multiplier n_{h}.

Implementation. populate_ion_matrix stores B in `ion_matrix`, despite that name suggesting a density-weighted rate. solve_no_source and solve_source multiply by the electron-density grid before invoking the matrix-exponential solvers. Do not apply the density multiplier a second time when interpreting their populations.

13.1 Ionization age

For constant electron density, define $\tau = n_{\mathrm{e}} t$. If B has no density dependence, a solution at fixed temperature can be written as $x(\tau) = \exp(B\tau)x(0)$. In a full GCR model, quenching and three-body terms can make B density-dependent, so equal ionization age does not necessarily imply equal charge-state fractions at different densities.

Ionization-age grids. The wrapper constructs times as $t = \tau/n_{\mathrm{e}}$ with axes *density, time* when there are multiple densities; one density uses a one-dimensional time array. This agrees with the matrix-exponential solver's grid convention, including singleton axes.

13.2 Input routes and retained manifolds

ionization_balance can build coefficients from ADF04 models or read precomputed ADF11/HDF5 data. The number of retained metastables is a stage-dependent quantity. A ragged collection such as $[\,[0,1],\,[0,1,2],\,]$ must not be forced into a rectangular numeric array. The order of input files also carries charge-stage information and should be explicit in a calculation record. The current readers and constructors impose additional restrictions listed in Section 17.

Constructor defaults. For ADF04, a flat metastable array selects levels in the first stage; later selections use the preceding SCD parent count. A per-stage list may have unequal lengths. Default initial abundances are sized after the GCR stages are loaded, including the final parent stage, and place all population in the first state. ADF11 and HDF5 source solves use the same default-abundance helper after their ionization matrix is constructed.

13.3 Stage-to-stage coefficients

A stage-to-stage balance follows one population per charge stage, with the coefficients of conventional collisional-radiative models: SCD, ACD, CCD, PLT, PRB and PRC for each stage, and no QCD or XCD, which move population between the metastables of one stage or between parents [9]. They can be made from the GCR picture in two ways.

Retaining only the ground condenses the metastables onto the ground, which restores the ground-state collisional-radiative picture of Bates et al. [9]. Each metastable becomes an excited level whose excitation-driven population enters the stage normalization (Section 8.3) and the SCD, while its capture-driven population is not counted in the stage, and the ACD counts only captures that reach the ground.

Alternatively, the resolved coefficients are contracted with the metastable fractions $f_\rho = m_\rho / \sum_\sigma m_\sigma$ of the ionizing stage, and f_p^+ of the recombining stage, as ADAS404 makes its

unresolved ADF11 files (Section 15.3):

$$\mathcal{S}_{\text{stage}} = \sum_{\rho} f_\rho \sum_{p} \mathcal{S}_{\rho p}, \qquad \text{PLT}_{\text{stage}} = \sum_{\rho} f_\rho \, \text{PLT}_\rho, \qquad \mathcal{A}_{\text{stage}} = \sum_{p} f_p^+ \sum_{\rho} \mathcal{A}_{\rho p}. \tag{13.2}$$

The fractions tell how the population of a stage is shared among its metastables, which a single population per stage cannot track. The ratios m_ρ to the ground ($m_g = 1$), the MET data of ADAS, are the steady state of the retained metastables of the stage alone: excitation, de-excitation and their ionization loss, without capture from the next stage. For a ground g and one metastable μ,

$$m_\mu = \frac{\mathcal{Q}_{g\to\mu}}{\mathcal{Q}_{\mu\to g} + \sum_p \mathcal{S}_{\mu p}}, \tag{13.3}$$

and a linear system for more metastables. In a plasma the metastable is also fed by capture, and in steady state

$$\frac{N_\mu}{N_g} = m_\mu + \frac{\sum_p \mathcal{A}_{\mu p} N_p^+}{(\mathcal{Q}_{\mu\to g} + \sum_p \mathcal{S}_{\mu p})\, N_g}. \tag{13.4}$$

The contraction neglects the second term, which holds in an ionizing plasma, where the next stage is scarce. Where the next stage dominates, most metastable atoms come from capture and are ionized again before they reach the ground. Equation (13.2) then weights the ionization with too small a metastable share; metastables ionize more readily than the ground, so the stage-to-stage SCD is too small and the population of the stage too large. The condensation onto the ground errs the other way: it leaves the capture-fed metastable atoms out of the stage, whose population from ACD/SCD is then too small. Retaining the metastables avoids both assumptions. Summers et al. also note that XCD is needed to build a meaningful stage-to-stage ionization coefficient from the GCR ones [9]: it moves population between the metastables of the next stage through this one, and the contraction leaves it out of the fractions of that stage, where it would add to its QCD.

ionization_balance makes the contraction from ADF04 inputs with `use_met=True` (default `False`): each stage is solved with its metastables, metastable_fractions gives their fractions from its QCD and SCD, Eq. (13.2) gives one SCD, ACD, CCD and PLT per stage, and the balance is solved stage to stage. It reproduces ADAS's unresolved data where ADAS's own data are consistent, but it is not more accurate than the condensation onto the ground. MET only counts metastable atoms made by excitation from the ground. In He^0, most recombinations land on the triplets, and the 1s2s ^{3}S metastable is easily ionized, so most of those atoms go straight back to He^+. The collapse counts them as recombined but never as re-ionized, so He^0 comes out too abundant. Be^0's ^{3}P holds more atoms than the ground, so there the condensation's error dominates instead: it drops the recombination-fed ^{3}P atoms from the stage entirely. Against the equilibrium $N(X^0)/N(X^+)$ of the resolved balance (1–100 eV, 10^{10}–10^{14} cm^{-3}, ADAS's ADF04, ADF17 and ADF18 files: `cpb03_ls#be0.dat` and `mom97_ls#he0.dat` with their `cbnm96` and `exp96` projection), the collapse gives 1.00–1.34 for Be^0 and 1.02–2.21 for He^0, the condensation onto the ground 0.26–1.00 and 0.88–1.00. So no stage-to-stage model is right for every ion. The physically correct option is the resolved balance, the default.

13.4 Electron energy loss, radiated power and emission functions

A fluid energy equation needs the energy that the free electrons lose to an ion. Summers et al. (their Eq. 27) write it from the GCR coefficients and the power coefficients,

$$L_e = \sum_X E_X \, \dot{N}_X\big|_{\text{atomic}} + n_e \sum_\rho \text{PLT}_\rho N_\rho + n_e \sum_p \text{PRB}_p N_p, \tag{13.5}$$

where X runs over the metastables of the ion and of its parent, E_X is the absolute energy of X, and $\dot{N}_X|_{\text{atomic}}$ its rate of change by the GCR coefficients: transfer between the metastables,

ionization and recombination. With PLT as in Eq. (10.6) it includes the forbidden lines between metastables. Only relative energies remain, because the atomic processes conserve particles, and in ionization equilibrium the first term vanishes: the electron energy loss equals the radiated power. For an element in ionization equilibrium the radiated power coefficient is (their Eq. 53)

$$P^{\mathrm{tot}} = \sum_z \sum_\sigma (\mathrm{PLT}_\sigma^{[z]} + \mathrm{PRB}_\sigma^{[z]}) \Big(\frac{N_\sigma^{[z]}}{N^{\mathrm{tot}}} \Big)_{\mathrm{eq}}, \tag{13.6}$$

in $\mathrm{W\,cm^3}$ per electron and per ion of the element, where $\mathrm{PRB}_\sigma^{[z]}$ is the recombination power driven by the metastable σ of stage z, which stage $z-1$ emits. The emission functions of a line, the GTN functions of Summers et al. (their Eq. 54, the $G(T_e)$ of solar physics at finite density), combine the PECs with the same fractions:

$$\mathrm{GTN}_{u\ell}^{[z]} = \sum_\sigma \mathrm{PEC}_{u\ell,\sigma}^{[z]\,\mathrm{exc}} \Big(\frac{N_\sigma^{[z]}}{N^{\mathrm{tot}}} \Big)_{\mathrm{eq}} + \sum_p \mathrm{PEC}_{u\ell,p}^{[z]\,\mathrm{rec}} \Big(\frac{N_p^{[z+1]}}{N^{\mathrm{tot}}} \Big)_{\mathrm{eq}}. \tag{13.7}$$

ColRadPy provides the coefficients and, from ionization_balance, the equilibrium metastable fractions (`processed['pops_ss']`), but does not assemble Eqs. (13.5)–(13.7). Its `prb` lacks the continuum and bremsstrahlung that the ADAS PRB supplies.

14 Charge-state energy balance

The energy-balance extension first solves the charge-state fractions $x_z(t)$, then evolves a kinetic-energy variable per modeled particle,

$$\mathcal{E}_z = \frac{3}{2} x_z T_z. \tag{14.1}$$

It transports that variable between charge states with the same ionization matrix and adds exchange with a prescribed background ion bath at T_b:

$$\dot{\mathcal{E}}_z = \sum_{z'} M_{zz'} \mathcal{E}_{z'} - \frac{2}{3} \nu_z \mathcal{E}_z + \nu_z x_z T_b. \tag{14.2}$$

The convention for ν_z is an energy-transfer frequency: $\frac{3}{2} x_z \dot{T}_z = -\nu_z x_z (T_z - T_b)$ when the population is fixed. The temperature-relaxation frequency is therefore $2\nu_z/3$. This is kinetic-temperature exchange, not a complete plasma energy equation including radiation, electron heating, and ionization-potential energy losses (Section 13.4).

14.1 Charged and neutral energy transfer

The charged helper implements

$$v_s^2 = 2eT_s/m_s, \tag{14.3}$$

$$\nu_{ab} = \frac{n_b Z_a^2 Z_b^2 e^4 \ln \Lambda}{\pi^{3/2} \epsilon_0^2 m_a m_b (v_a^2 + v_b^2)^{3/2}}. \tag{14.4}$$

In these expressions temperatures are numeric values in eV, eT is an SI energy, masses are in kg, and n_b is in m^{-3}. The helper evaluates

$$\lambda_D^{-2} = \sum_s \frac{n_s Z_s^2 e^2}{\epsilon_0 e T_s}, \tag{14.5}$$

$$b_{\mathrm{cl}} = \frac{|Z_a Z_b| e^2}{4\pi\epsilon_0 \mu (v_a^2 + v_b^2)/2}, \qquad b_{\mathrm{q}} = \frac{\hbar}{\mu\sqrt{v_a^2 + v_b^2}}, \tag{14.6}$$

$$\ln\Lambda = \ln \frac{\lambda_D}{\max(b_{\mathrm{cl}}, b_{\mathrm{q}})}, \qquad \mu = \frac{m_a m_b}{m_a + m_b}. \tag{14.7}$$

The screened, weakly coupled collision model is not reliable when its assumptions fail or its logarithm ceases to be physically meaningful.

For the neutral, the implemented polarization-scattering approximation uses

$$K_{\mathrm{mt}} = 2.21\pi\alpha_{\mathrm{f}} c \sqrt{\frac{m_e}{\mu} a_{\mathrm{pol}} a_0^4}, \qquad \nu_{nb} = \frac{3 n_b \mu}{m_n + m_b} K_{\mathrm{mt}}. \tag{14.8}$$

Here a_{pol} is the numeric polarizability in Bohr-radius cubed and α_{f} is the fine-structure constant. This elastic approximation does not include every long-range or inelastic energy-transfer mechanism.

Energy extension. The caller converts electron density from cm^{-3} to m^{-3} for the Debye helper. The energy solver uses one temperature/density point, reconstructs populations from stored eigenpairs, and integrates Eq. (14.2) with BDF. A failed integration gives a warning with the solver message, and the stored times, abundances, energies and temperatures stop at the last time reached.

15 Data products and external interfaces

15.1 Array layout

For an ordinary temperature–density grid, thermodynamic axes trail the physical indices: $(\ldots, N_T, N_n)$. Paired mode uses $(\ldots, N_{\mathrm{pair}})$, evaluating corresponding temperature and density entries rather than their Cartesian product. Typical arrays are

$$\begin{aligned}
\texttt{pops}: &\ (N_e, N_{\mathrm{driver}}, N_T, N_n),\\
\texttt{pecs}: &\ (N_{\mathrm{line}}, N_{\mathrm{driver}}, N_T, N_n),\\
\texttt{scd}, \texttt{acd}: &\ (N_m, N_p, N_T, N_n),\\
\texttt{qcd}: &\ (N_m, N_m, N_T, N_n),\\
\texttt{xcd}: &\ (N_p, N_p, N_T, N_n).
\end{aligned}$$

The paired layout removes one grid axis. An excited-level axis is not a physical bound-level index; use the stored level map. Likewise the driving column index combines several process types and does not universally mean an ionization parent. The index order of the effective coefficients is:

15.2 ADF04, ADF11, ADF13, ADF15 and HDF5

ADF04 is a level-resolved atomic-data format, whereas ADF11 holds effective charge-state coefficients on temperature and density grids, made as described in Section 15.3. The ADF11 reader stores logarithmic data and the ionization-balance path interpolates the logged grids and coefficients before converting to physical values. File naming and block interpretation identify

Array	First index	Second index	Rate multiplier
`scd`	source bound metastable	destination parent	n_e
`acd`	destination bound metastable	source parent	n_e
`ccd`	destination bound metastable	source CX parent	n_h
`qcd`	source bound metastable	destination bound metastable	n_e
`xcd`	source parent	destination parent	n_e

Table 4: Stored coefficient orientation. Equal matrix dimensions can conceal an accidental transpose, so checking shapes alone is insufficient.

coefficient families. ADF13 holds S/XB ratios of selected lines in blocks like those of ADF15, each with its driver [10]; ColRadPy reads them and writes its own from the SCD and excitation PECs (Section 10.3).

ADF15 records line photon emission coefficients with wavelength and driver identifiers. Its wavelength convention is Angstroms [11], so a conversion from internal nm must occur exactly once. The number of output blocks must include every emitted line–driver pair; each block needs an unambiguous index and process/driver metadata. These are part of the scientific interface because a downstream reader can silently drop or misidentify lines.

Export and reading. The ADF15 wrapper converts nm to Angstroms once; the writer declares every line–driver block and assigns distinct selectors, and writes the ion field (`C + 2`) and a source member name, without which the ADAS reader stops. The ADF13 writer uses the same fields, with `INDM = T` for populations normalized to the whole ion and the driver index otherwise; the ADAS reader keeps three digits of the selector, so at most 999 blocks are read. The ADF11 constructor collects coefficient files before filling missing QCD/XCD arrays, so their order does not require SCD first. The readers follow the fixed fields of the ADAS specifications: ADF04 values are 8-character fields, so a negative value can follow the previous one without a space; ADF11 blocks are placed by their labels (IPRT, IGRD, JPRT, JGRD), whose order and meaning change with the class, including metastable resolved power files; each ADF13 block has its own grids and metastable. They were checked against the ADAS data collections: every ADF11 and ADF13 file, the condensed projection ADF17 files and a sample of ADF04 files from every collection.

The HDF5 dumping method uses an optional serialization dependency and can select PECs, GCR coefficients, and additional arrays. Record the data version, source file and hash, selected metastables, units, grid ordering, normalization convention, and any rate supplementation with each saved result.

15.3 How ADAS makes its ADF11 data

ADF11 is a derived format [5]: its coefficients, functions of the electron temperature and density, come from collisional-radiative models and are given by element at two levels of refinement [9]. In the *unresolved* or stage-to-stage data only the ground states of the ions are taken as dominant populations: these are conventional collisional-radiative (CR) coefficients, one per charge stage, of the classes ACD, SCD, CCD, PLT, PRB and PRC (for example `scd96`, `acd96`). In the *metastable resolved* data the ground and the metastables of each ion may be dominant: these are the generalized collisional-radiative (GCR) coefficients of Section 9, the same classes with the block of each ion subdivided by metastable (and parent), plus QCD and XCD (for example `scd96r`, `qcd96r`, `xcd96r`). QCD and XCD exist only for GCR data, because they move population between the metastables of one stage or between parents, which a single population per stage does not have. Three codes make the GCR-project data (classes `96`). ADAS204 [6] solves a bundle-nS model and writes the condensed matrices that Section 7 projects onto the low levels.

ADAS208 [7] solves the model of an ADF04 file with that projection and partitions the reduced matrix into the coefficients above: the ionization of each level is that of the S lines of the file, or ECIP without them, optionally replaced by zero-density ADF07 rates for the metastables, and three-body recombination follows from it by detailed balance, as in ColRadPy. The results are stored per isoelectronic sequence (ADF10 [4]) on the scaled grids T_e/z_1^2 and n_e/z_1^7, z_1 being the charge of the recombining ion, and ADAS404 [8] makes the ADF11 files by cubic-spline interpolation in z_1, temperature and density. PRB does not come from ADAS208: it is the bundle-n recombination power of ADAS204, the recombination continuum, bremsstrahlung weighted by spin system, the cascade and dielectronic stabilization, whereas `prb` in ColRadPy is the line power of the levels of the ADF04 file after capture only.

The unresolved coefficients of the GCR project (classes `96`, and `93` from the earlier resolved data), although described as conventional CR data, are not made with a separate model: ADAS404 contracts the resolved ones with metastable fractions, Eq. (13.2). The contraction changes only the stages with more than one metastable: in the `96r` files these are, for example, He^0, Li^+, Be^0 and Be^{2+}, and all but the H-, Li- and Na-like stages of C, N, O, Ne and Si. The fractions are an ADF10 class of their own (MET, "equilibrium metastable fractions", `adf10/met96/pj#met96_<seq>##.dat`, written by ADAS208 for each ion), which holds the ratios $m_\rho = N_\rho/N_g$ to the ground (for Be^0 $m_{3P} = 1.49$ at 2 eV) and which ADAS404 normalizes to fractions after interpolation. Other unresolved classes (`74`, `85`, `89`, `92`, `12`, `50`) come from other codes without resolved data; the code that made a file is named in its comments.

For line emission "unresolved" means the condensation onto the ground instead: ADAS's unresolved photon emissivity and S/XB files (prefixes `pju`, `llu`; `INDM = T`) come from an ADAS208 run with the ground as the only metastable and the populations normalized to the whole stage, its resolved ones (`pjr`, `llr`) from a run with the metastables. ColRadPy reproduces the former with the ground as its only metastable.

16 Analytic examples and checks

16.1 A two-level density scan

For a two-level ion with upward and downward coefficients q_{01}, q_{10} and radiative rate A, the stationary, ion-total-normalized excited fraction is

$$f_1 = \frac{n_e q_{01}}{A + n_e(q_{01} + q_{10})}, \qquad \mathrm{PEC}_{10} = \frac{A q_{01}}{A + n_e(q_{01} + q_{10})}. \tag{16.1}$$

At low density, PEC approaches q_{01} and f_1 is proportional to density. At high density, $f_1 \to q_{01}/(q_{01}+q_{10})$ while PEC falls inversely with density. The photon rate per emitting ion approaches Af_1, not zero. This distinction explains why a falling PEC is not itself vanishing emission.

16.2 A conservative chain and a slow mode

The generator in Eq. (12.5) is closed and has a known exact solution. An independent SciPy matrix exponential reproduces it; the ColRadPy solver agrees with both it and the analytic solution while conserving population.

A second generator, $C = [[-r, 0], [r, 0]]$ with $r = 10^{-12}\,\mathrm{s}^{-1}$, has $N_0(10^{12}\,\mathrm{s}) = \mathrm{e}^{-1}$ for $N(0) = (1, 0)$. The source solver with a zero source reproduces this slow evolution and agrees with source-free dynamics.

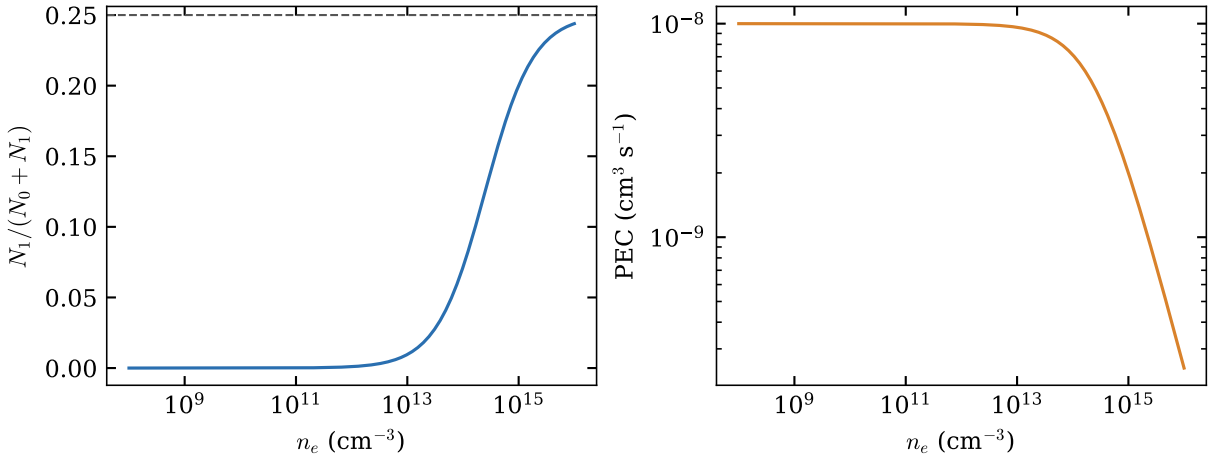


Figure 2: An analytic illustration with $A = 10^7\,\mathrm{s}^{-1}$, $q_{01} = 10^{-8}$ and $q_{10} = 3 \times 10^{-8}$ cm^3/s. The dashed line is the high-density fraction 1/4. These are illustrative coefficients, not a fit to a specific atom or an execution of the ColRadPy atomic-data path.

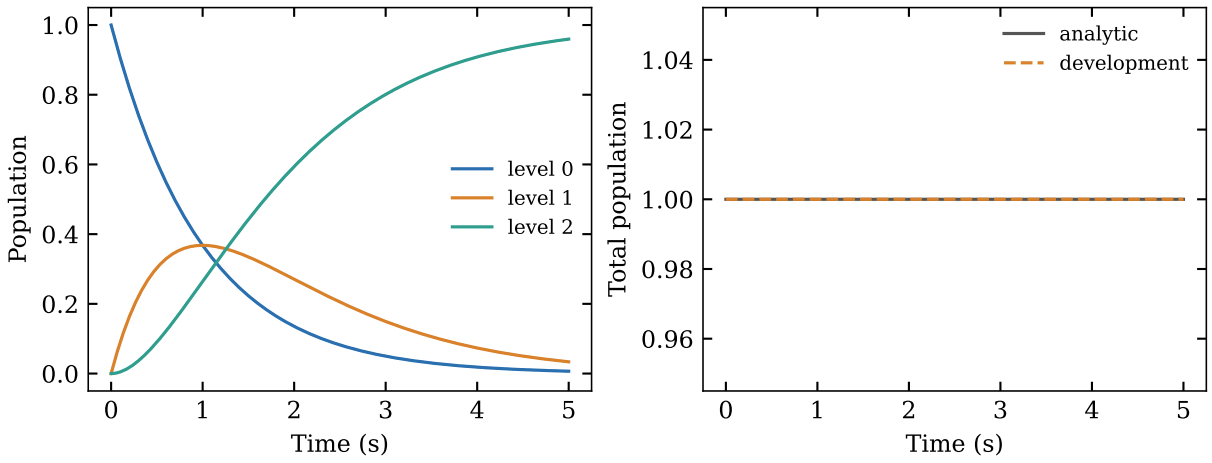


Figure 3: Equal-rate chain: solid curves are analytic populations; dashed curves are ColRadPy results. The right panel tests the conserved sum. Agreement includes the te^{-t} contribution from the defective basis.

16.3 XCD checked by direct elimination

The bundled Be II file is solved with metastables $[0, 1]$, temperatures $[5, 20]$ eV and densities $[10^{12}, 10^{14}]$ cm^{-3}, with both recombination mechanisms and ionization enabled. At the first grid point, form the parent block of Eq. (8.3) using a linear solve rather than the stored inverse. Divide its off-diagonal rate by n_e and compare with stored XCD$[1, 0]$. Both quantities are in cm^3/s; a separate noncontiguous-metastable test covers row selection.

Quantity	Development	Reference
Chain: $N_1(1)$	0.367879	0.367879
Slow mode: $N_0(10^{12})$	0.367879	0.367879
Be II: XCD$[1, 0]$	3.8811×10^{-12}	3.8811×10^{-12}

Table 5: Executed checks. The first two rows are dimensionless populations. Both XCD values have units cm^3/s.

The figure script records the branch HEAD, working-tree status, package-module hashes, Python and library versions in `provenance.json`. Grid parameters and numerical results are recorded in `checks.json`. The analytic figures and checks establish specific limits and agreements. They are not a certification of all atomic datasets, an ADAS benchmark, or a reproduction of a particular tungsten dataset.

17 Validity boundaries and implementation checklist

17.1 Physical assumptions

The model assumes a specified Maxwellian electron bath, rates appropriate to the supplied atomic resolution, and escaping photons. The quasistatic approximation additionally requires fast excited-state relaxation. Statistical LS multiplet splitting assumes the relevant coupling and population pattern. ECIP approximates missing ionization; extrapolation does not create new atomic accuracy. The energy extension describes trace species kinetics in a prescribed background. These assumptions are separate from implementation defects.

17.2 Current limitations

Area	Current qualification	Discussed in
Projection	The fractional-parentage switches of the `cbnm93` ADF17 files and indirect charge exchange are not implemented; the ACD into Ne^+ differs from ADAS by 20–40 %; below the first ADF17 temperature the couplings are continued with estimated thresholds and limited power laws, and above the last one held at its values.	Section 7

Area	Current qualification	Discussed in
Radiated power	`prb` is the line power of the ADF04 levels after capture; it does not include the recombination continuum, bremsstrahlung or the radiation of the projected shells that the ADAS PRB contains (1–8 % of it for Be I).	Section 15.3
Processes not modelled	Ionization from the stage below into the metastables (the ADF04 I lines; the ionization from the stage below of the GCR equations of Summers et al.), proton and other ion impact (P lines), and unresolved dielectronic satellite lines and the autoionization of levels of the ADF04 file (L lines and the AAUG values) are not modelled; read_adf04 skips the P, I and L lines with a warning. The electron energy loss, the radiated power of an element and GTN functions are not assembled.	Sections 9, 13.4
Coupling assumptions	Intercombination and $\|\Delta L\| > 1$ lines are kept unsplit rather than split with a coupling model.	Section 11

The table is a guide to interpreting the current implementation. Before using a result, check coefficient signs, state and parent maps, column-sum conservation, population normalization, line-energy consistency, density powers, and the exported units. For a transient solve, compare against an independent matrix exponential at representative grid points and verify the initial condition. For a quasistatic solve, substitute the calculated excited populations back into the balance equation. Agreement with these identities does not establish the accuracy of the input atomic calculations, but a failure identifies a problem before comparison with an experiment.

A Source-code map

The map below is generated from the same source as the numerical checks. Paths are relative to the repository root.

File	Entry point	Line
`colradpy/colradpy_class.py`	populate_data	641
same	make_ioniz_from_reduced_ionizrates	816
same	suppliment_with_ecip	893
same	make_cx_rates_from_file	956
same	make_recombination_rates_from_file	999
same	make_three_body_recombination	1073
same	make_electron_excitation_rates	1118
same	populate_cr_matrix	1280
same	solve_quasi_static	1601
same	solve_time_dependent	2048
same	split_pec_multiplet	2147
same	shift_j_res_energy_to_nist	2564
same	solve_cr	2700
same	make_nat_broad	2742

File	Entry point	Line
same	make_dopp_broad	2820
same	write_pecs_adf15	3146
same	get_sxb	3206
same	write_sxb_adf13	3242
colradpy/ionization_balance_class.py	populate_ion_matrix	531
same	solve_no_source	679
same	solve_source	717
same	solve_time_independent	782
same	metastable_fractions	44
same	collapse_metastables	490
colradpy/solve_matrix_exponential.py	solve_matrix_exponential	245
same	eval_matrix_exponential_solution	318
same	solve_matrix_exponential_steady_state	349
same	solve_matrix_exponential_source	442
colradpy/energy_balance.py	solve	88
same	plasma_energy_transfer_frequency	243
same	neutral_energy_transfer_frequency	298
same	coulomb_logarithm	351
same	debye_length	393
colradpy/read_fac.py	read_fac	642
same	_fold_autoionization	595
same	fac_cross_sections	505
colradpy/convolve_eedf.py	rate_coefficients	68
same	resonance_rate_coefficients	105
same	maxwellian	29

References

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[2] ADAS, *OPEN-ADAS report 1*, especially the generalized collisional radiative definitions and separate emission drivers. https://www.adas.ac.uk/adaseu_site/reports/open_1.pdf.

[3] ADAS User Manual, Appendix A, *ADF04: resolved specific ion data collections*. https://www.adas.ac.uk/man/appxa-04.pdf.

[4] ADAS User Manual, Appendix A, *ADF10: iso-electronic master files*. https://www.adas.ac.uk/man/appxa-10.pdf.

[5] ADAS User Manual, Appendix A, *ADF11: iso-nuclear master files*. https://www.adas.ac.uk/man/appxa-11.pdf.

[6] ADAS User Manual, *ADAS204: specific z recom/ionis – process collisional radiative coefficients and populations*. https://www.adas.ac.uk/man/chap2-04.pdf.

[7] ADAS User Manual, *ADAS208: specific z excitation – advanced population processing*. https://www.adas.ac.uk/man/chap2-08.pdf.

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[10] ADAS User Manual, Appendix A, *ADF13: ionisation per photon coefficients*. https://www.adas.ac.uk/man/appxa-13.pdf.

[11] ADAS User Manual, Appendix A, *ADF15: photon emissivity coefficients*. https://www.adas.ac.uk/man/appxa-15.pdf.

[12] ColRadPy development branch working tree used for this manual. The recorded HEAD, source-file hashes and runtime versions are in `docs/source/theory/figures/provenance.json`.

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